

Adam Shaw

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EDUCATION

University of Southern California

Los Angeles, CA

M.S. in Statistics

Expected December 2027

B.S. in Applied and Computational Mathematics, Minor in Computer Science

Expected May 2027

Undergraduate GPA 3.7 · USC Presidential Scholar

University of Edinburgh Visiting Student

Spring 2026

PUBLICATIONS

- [1] **Adam Shaw**, Yixuan Sun, Srinivas Eswar, William Detmold, Phiala Shanahan, Xiaoye Li, and Yang Liu. “Matrix-Free Gauge-Equivariant Neural Preconditioners for the Wilson–Dirac Operator.” *Preprint, 2026*. To be submitted to the NeurIPS 2026 Workshop on Representations for the Physical Sciences.
- [2] **Adam Shaw**, Jiayu Li, Michael Sperling, Michael Kim, and Alvin Jin. “**Topological Simplification in Predictive Coding Networks**.” *Proceedings of the 2nd Conference on Topology, Algebra, and Geometry in Data Science (TAG-DS 2026)*, PMLR 334(2):316–332, 2026.
- [3] **Adam Shaw**, Jiayu Li, Michael Sperling, Michael Kim, and Alvin Jin. “Symmetry Acquisition in Predictive Coding Networks.” *ICML 2026 Workshop on Weight-Space Symmetries*. Accepted (non-archival).

RESEARCH EXPERIENCE

Argonne National Laboratory (SULI Intern)

Summer 2026

Gauge-equivariant neural preconditioners for lattice QCD

Lemont, IL

- Designed and ran a study of gauge-equivariant neural preconditioners for solving the Dirac equation in lattice QCD, evaluating model architectures and loss functions against iterations to convergence in conjugate gradient.
- Delivered a model that cut solver iterations roughly 2× below the state-of-the-art method while reducing its per-system setup cost.
- Proposed three novel architectures and diagnosed why they failed to improve results, tracing the outcome to specific structural properties and narrowing where future gains are likely to be found.
- Built the full training and evaluation pipeline in JAX, including automatic differentiation through an iterative solver and multi-GPU data-parallel training.
- First-authored the resulting paper with collaborators at MIT and Lawrence Berkeley National Laboratory.

University of Southern California

Fall 2025 – Spring 2026

Topological deep learning, advised by Prof. Alvin Jin

Los Angeles, CA

- Analyzed how topological invariants (Betti numbers) change across network layers and how those changes affect inversion ability in predictive coding networks, a type of invertible neural network.

- Developed an extensive Python codebase implementing predictive coding networks in JAX and topological analysis with Ripser, NumPy, and scikit-learn.
- Produced two accepted papers — TAG-DS 2026 and the ICML 2026 Workshop on Weight-Space Symmetries — and a poster at the Conference on Topological Data Analysis, University of Missouri.

POSTERS

<i>Matrix-Free Gauge-Equivariant Neural Preconditioners for the Wilson-Dirac Operator.</i>	SULI Poster Session, Argonne National Laboratory.	Aug. 2026
<i>Topological Simplification in Predictive Coding Networks.</i>	Conference on Topological Data Analysis, University of Missouri.	Nov. 2025

INDUSTRY EXPERIENCE

Google — Associate Software Developer Intern	Summer 2025
<i>Workspace AI Platform — Gemini integration across Docs, Slides, and Gmail</i>	New York, NY
• Designed and built an access control system for custom Gems (personalized Gemini instances), streamlining creation and secure sharing across Google Workspace teams. • Implemented role-based permissions in C++ on Google’s internal server framework, handling authorization and routing for millions of potential users. • Wrote a Python automation script to manage enum ID generation and updates across a 10,000+ line shared codebase, eliminating merge conflicts and synchronization errors for a multi-engineer team.	
Rollick — Software Engineer Intern	Summer 2024
<i>Front-end development and campaign personalization</i>	Chicago, IL
• Built modular HTML templates in Nunjucks and added dynamic interactivity with jQuery; tested and ran code through Node.js and npm scripts. • Optimized UTM parameters to identify users more effectively and alter the product accordingly. • Worked in a KanPlan (Jira) workflow with per-issue Git branches and GitHub pull requests reviewed by senior developers.	

AWARDS AND HONORS

USC Presidential Scholar	2023
National Merit Scholarship Winner	2023

RELEVANT COURSEWORK

Graduate	Applied Probability, Analysis of Algorithms
Mathematics	Probability Theory, Numerical Methods, Mathematics of Machine Learning, Linear Algebra, Mathematical Statistics, Real Analysis, Number Theory
Computer Science	Natural Language Processing, Data Science with Python, Discrete Mathematics, Data Structures, Introduction to Programming (C++)